THRUST BEARINGS

●Thrust Needle Roller Bearings
●Thrust Roller Bearings

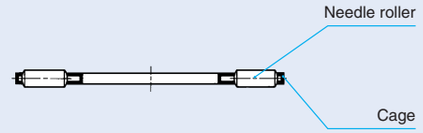


Structure and Features

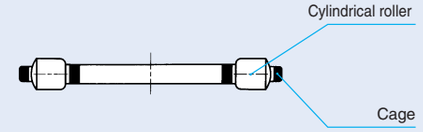
IKO Thrust Bearings consist of a precisely made cage and rollers. They have high rigidity and high load capacities and can be used in small spaces.
Thrust Needle Roller Bearings incorporate needle rollers, while Thrust Roller Bearings incorporate cylindrical rollers. Various types of raceway rings are available, and suitable bearings can be selected according to the operating conditions.
When the bearing mounting surfaces of a machine are heat-treated and finished by grinding as raceways, Thrust Bearings can be used without raceway rings allowing the machine to be made more compact. They are most suited to applications where high accuracy is required at high speeds and under fluctuating heavy loads, such as driving mechanisms for automobiles, machine tools, and high-pressure pumps.

Structures of Thrust Bearings

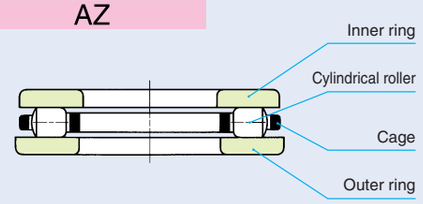
NTB



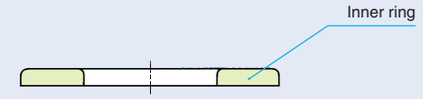
AZK



AZ



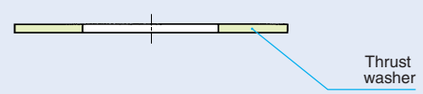
WS



GS



AS



Types

In IKO Thrust Bearings, the types shown in Table 1 are available.

Table 1.1 Type of bearing

Type	Thrust needle roller bearings	Thrust roller bearings	
		Without inner and outer rings	With inner and outer rings
Model code	NTB	AZK	AZ

Table 1.2 Type of bearing ring

Type	Inner ring	Outer ring	Thrust washer
Model code	WS	GS	AS

Thrust Needle Roller Bearings

These bearings consist of a cage made from a steel plate, which is precisely press formed and surface-hardened, and needle rollers with a diameter variation within 2 μm. They have a rigid structure and a high lubricant-retaining capacity.
As they have the lowest sectional height compared with other thrust bearings, they can be used instead of conventional thrust washers and can withstand high-speed rotations with a low coefficient of friction.
Specially designed thin inner rings (WS) and outer rings (GS), and especially thin (1 mm thick) thrust washers (AS), are available for use in various applications.
These bearings are generally used by utilizing their inner surface as the guide surface.

Thrust Roller Bearings

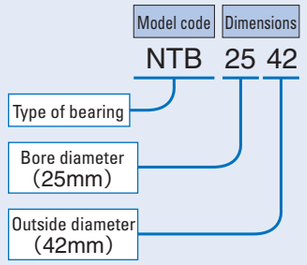
In this series, the caged cylindrical rollers AZK and the complete bearings AZ in which AZK are combined with an inner ring (WS) and an outer ring (GS) are available.
The cage has a special precise structure which is highly rigid, and cylindrical rollers are outwardly arranged and guided by the cage with exact precision to enable them to withstand heavy loads even at high rotational speeds.
Owing to the high accuracy of the bearing height T, they are suitable for use in machine tools, ultra-high pressure pumps, etc.
These bearings are generally used by utilizing their inner surface as the guide surface.

Identification Number

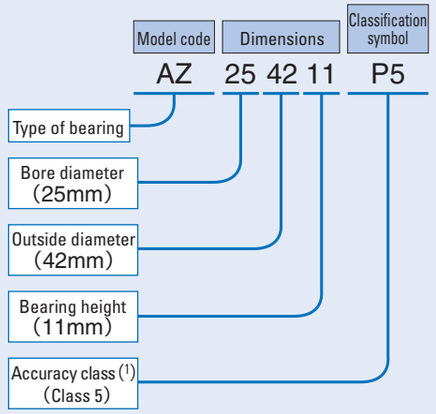
The identification number of Thrust Bearings consists of a model code, dimensions and a classification symbol. Some examples are shown below.

Examples of identification number

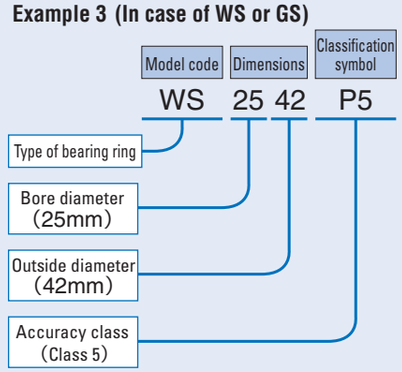
Example 1 (In case of NTB or AS)



Example 2 (In case of AZ or AZK)



Example 3 (In case of WS or GS)



Note(1) Not applicable to the model AZK.

Accuracy

The accuracy of Thrust Bearings is based on JIS B 1514:2000 as shown in Table 2.

Table 2.1 Tolerances

unit: μm

Type of bearing \ Item		Dimension	Dimension symbol	Tolerance	
Thrust needle roller bearings	NTB	Bore diameter	d	E11	
		Outside diameter	D	c12	
		Width	D_w	Equivalent to JIS B 1506 Class 2	
Thrust roller bearings	AZK	Bore diameter	d_c	As per Table 2.2	
		Outside diameter	D_c		
		Width	D_w	$1 \leqq D_w \leqq 10$	Equivalent to JIS B 1506 Class 2
				$10 < D_w \leqq 30$	Equivalent to JIS B 1506 Class 3
	AZ	Height	T	As per Table 2.3	
Inner rings	WS	Bore diameter	d	As per Table 2.4	
		Outside diameter	D	b12	
		Width	B	h11	
Outer rings	GS	Bore diameter	d	B12	
		Outside diameter	D	As per Table 2.4	
		Width	B	h11	
Thrust washers	AS	Bore diameter	d	E12	
		Outside diameter	D	e12	
		Width	s	$\pm$50	

Table 2.2 Tolerances of bore and outside diameters for AZK series

unit: μm

Nominal dimension mm		Δ_{dc} Cage bore diameter deviation		Δ_{Dc} Cage outside diameter deviation	
Over	Incl.	High	Low	High	Low
—	50	+100	0	0	− 300
50	100	+200	0	0	− 400
100	200	+300	0	0	− 500
200	300	+500	0	0	− 700
300	400	+700	0	0	−1000
400	500	—	—	0	−1200

Table 2.3 Tolerances of height for AZ series

unit: μm

d Nominal bearing bore dia. mm		Δ_{Ts} Deviation of an actual bearing height	
Over	Incl.	High	Low
—	18	0	− 75
18	30	0	− 75
30	50	0	−100
50	80	0	−125
80	120	0	−150
120	180	0	−175
180	250	0	−200
250	315	0	−225
315	400	0	−300
400	500	0	−400

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

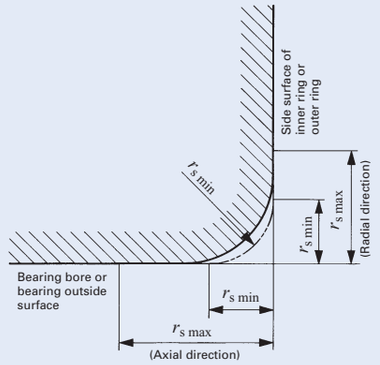
Table 2.4 Tolerances and allowable values for WS and GS

unit: μm

d or D (1) Nominal bearing bore dia. or outside dia. mm		Inner ring			Outer ring			Inner ring or outer ring		
		Δ_{dmp} Single plane mean bore diameter deviation		V_{dsp} Bore diameter variation in a single radial plane	Δ_{Dmp} Single plane mean outside diameter deviation		V_{Dsp} Outside diameter variation in a single radial plane	S_i or S_e (2) Bearing ring thickness variation		
								Class 0	Class 6	Class 5
Over	Incl.	High	Low	Max.	High	Low	Max.	Max.		
—	18	0	−8	6	0	−11	8	10	5	3
18	30	0	−10	8	0	−13	10	10	5	3
30	50	0	−12	9	0	−16	12	10	6	3
50	80	0	−15	11	0	−19	14	10	7	4
80	120	0	−20	15	0	−22	17	15	8	4
120	180	0	−25	19	0	−25	19	15	9	5
180	250	0	−30	23	0	−30	23	20	10	5
250	315	0	−35	26	0	−35	26	25	13	7
315	400	0	−40	30	0	−40	30	30	15	7
400	500	0	−45	34	0	−45	34	30	18	9

Notes(1) d for Δ_{dmp} and V_{dp}, and D for Δ_{Dmp} and V_{Dp}, respectively.
d for thickness variations of inner and outer rings .
(2) d_i for thickness variations of rings for NAX(I) and NBX(I) .

Table 2.5 Permissible limit values for chamfer dimension



unit: mm

$r_{s\,min}$	Radial and axial directions $r_{s\,max}$
0.3	0.8
0.6	1.5
1	2.2
1.1	2.7
1.5	3.5
2	4
2.1	4.5
3	5.5
4	6.5
5	8

Fit

The recommended fits for Thrust Bearings are shown in Table 3.

Table 3 Recommended fits

Type of bearing		Tolerance class	
		Shaft	Housing bore
Thrust needle roller bearings	NTB	h8(h10)	——
Thrust roller bearings	AZK	h6(h8)	——
	AZ		H7(H9)
Inner rings	WS	h6(h8)	——
Outer rings	GS	——	H7(H9)
Thrust washers	AS	h8(h10)	——

Mounting

When mounting Thrust Bearings, the following items should be considered.

❶When inner and outer rings are not used, the hardness of the raceway surfaces should be 58～64HRC, the effective hardening depth should be adequate, and the surface roughness should be less than 0.2 μm R_a.

❷When mounting inner and outer rings to shaft and housing bore, dimensions related to mounting should be based on the dimension tables.
Also, the mounting surfaces should be finished at right angles to the center axis and they should be sufficiently rigid.

❸To avoid elastic deformation, the thrust washer AS must be seated uniformly on its mating surface.
A small warp in an AS washer will be corrected automatically when an axial load is applied.

❹Thrust Roller Bearings are combinations of a copper alloy component and cylindrical rollers. When handling the AZK itself, care should be taken to prevent deformations, blemishes, etc.

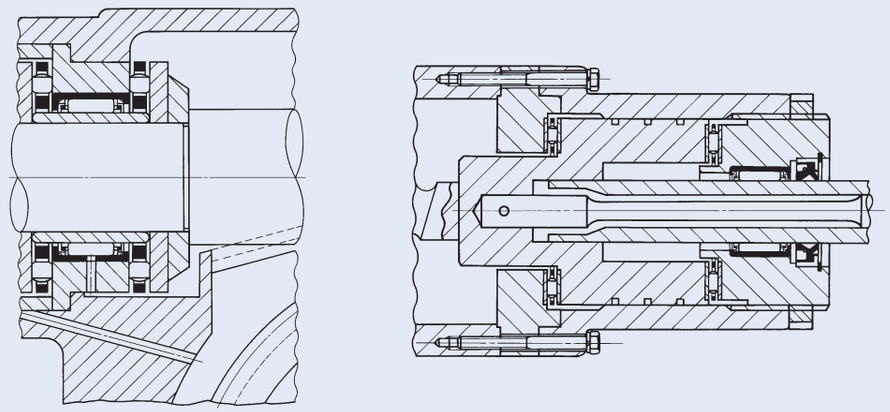


Fig.1 Mounting examples

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

THRUST BEARINGS

Thrust Needle Roller Bearings

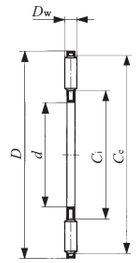




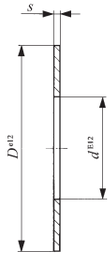
Shaft dia. 10 — 85mm

Shaft dia. mm	Identification number: Thrust needle roller bearing	Mass (Ref.) g	Thrust washer	Mass (Ref.) g	Inner ring	Outer ring	Mass (Ref.) g
10	NTB 1024	3.3	AS 1024	2.9	WS 1024	GS 1024	8
12	NTB 1226	3.8	AS 1226	3.2	WS 1226	GS 1226	8.9
15	NTB 1528	4.1	AS 1528	3.4	WS 1528	GS 1528	9.3
16	NTB 1629	4.3	AS 1629	3.6	WS 1629	GS 1629	9.8
17	NTB 1730	4.5	AS 1730	3.7	WS 1730	GS 1730	10.2
18	NTB 1831	4.7	AS 1831	3.9	WS 1831	GS 1831	10.7
20	NTB 2035	6.1	AS 2035	5	WS 2035	GS 2035	13.8
25	NTB 2542	8.2	AS 2542	6.9	WS 2542	GS 2542	21
30	NTB 3047	9.4	AS 3047	7.9	WS 3047	GS 3047	24
35	NTB 3552	10.6	AS 3552	8.9	WS 3552	GS 3552	31.5
40	NTB 40603	22	AS 4060	12.1	WS 4060	GS 4060	42.5
45	NTB 4565	24.5	AS 4565	13.3	WS 4565	GS 4565	53.5
50	NTB 5070	26.5	AS 5070	14.5	WS 5070	GS 5070	58.5
55	NTB 5578	33.5	AS 5578	18.5	WS 5578	GS 5578	93
60	NTB 6085	38.5	AS 6085	22	WS 6085	GS 6085	105
65	NTB 6590	41.5	AS 6590	23.5	WS 6590	GS 6590	124
70	NTB 7095	61	AS 7095	25	WS 7095	GS 7095	132
75	NTB 75100	65	AS 75100	26.5	WS 75100	GS 75100	153
80	NTB 80105	68.5	AS 80105	28	WS 80105	GS 80105	162
85	NTB 85110	72	AS 85110	29.5	WS 85110	GS 85110	170

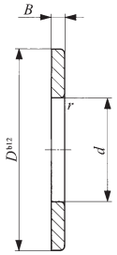
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 25% of this value is allowable.



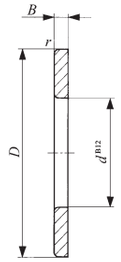
NTB



AS



WS



GS

Boundary dimensions mm d	D	D_w	s	B	$r_{s\,min}$ (1)	C_i	C_e	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
10	24	2	1	2.75	0.3	14	22	7 820	23 900	15 000
12	26	2	1	2.75	0.3	16	24	8 340	26 900	13 000
15	28	2	1	2.75	0.3	18	26	8 830	29 900	12 000
16	29	2	1	2.75	0.3	19	27	9 070	31 400	11 000
17	30	2	1	2.75	0.3	20	28	9 320	32 900	11 000
18	31	2	1	2.75	0.3	21	29	9 550	34 400	10 000
20	35	2	1	2.75	0.3	23	33	11 700	46 500	9 000
25	42	2	1	3	0.6	29	40	14 400	64 700	7 500
30	47	2	1	3	0.6	34	45	15 400	73 300	6 500
35	52	2	1	3.5	0.6	39	50	16 300	81 900	5 500
40	60	3	1	3.5	0.6	45	57	24 200	108 000	5 000
45	65	3	1	4	0.6	50	62	25 900	121 000	4 500
50	70	3	1	4	0.6	55	67	27 600	135 000	4 000
55	78	3	1	5	0.6	61	75	32 400	171 000	4 000
60	85	3	1	4.75	1	66	82	38 200	219 000	3 500
65	90	3	1	5.25	1	71	87	40 100	237 000	3 000
70	95	4	1	5.25	1	75	91	47 400	244 000	3 000
75	100	4	1	5.75	1	80	96	48 400	256 000	3 000
80	105	4	1	5.75	1	85	101	49 500	267 000	2 500
85	110	4	1	5.75	1	90	106	50 300	279 000	2 500

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

THRUST BEARINGS

Thrust Needle Roller Bearings

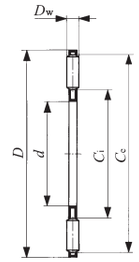




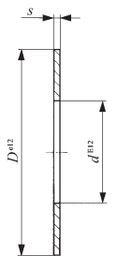
Shaft dia. 90 — 130mm

Shaft dia. mm	Identification number						
	Thrust needle roller bearing	Mass (Ref.) g	Thrust washer	Mass (Ref.) g	Inner ring	Outer ring	Mass (Ref.) g
90	**NTB 90120**	92	**AS 90120**	38	**WS 90120**	**GS 90120**	250
100	**NTB 100135**	119	**AS 100135**	50	**WS 100135**	**GS 100135**	350
110	**NTB 110145**	129	—	—	**WS 110145**	**GS 110145**	380
120	**NTB 120155**	139	—	—	**WS 120155**	**GS 120155**	410
130	**NTB 130170**	225	—	—	**WS 130170**	**GS 130170**	660

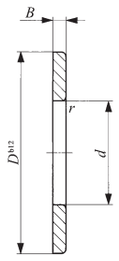
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 25% of this value is allowable.



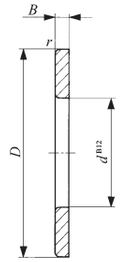
NTB



AS



WS



GS

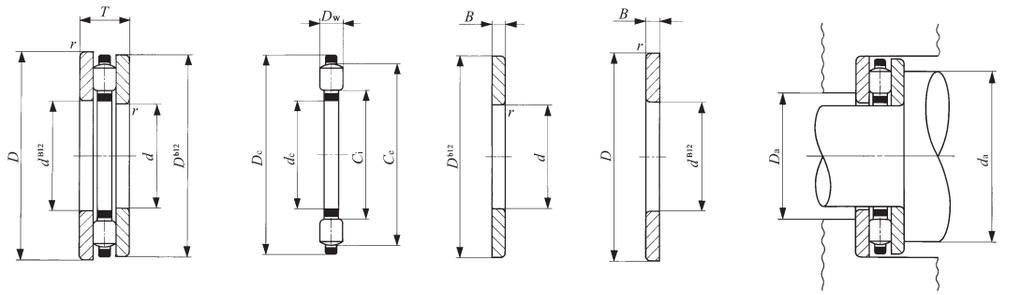
Boundary dimensions mm								Basic dynamic load rating C	Basic static load rating C_0	Allowable rotational speed(2)
d	D	D_w	s	B	$r_{s\,min}$ (1)	C_i	C_e	N	N	rpm
90	120	4	1	6.5	1	96	116	64 500	394 000	2 500
100	135	4	1	7	1	107	131	80 300	541 000	2 000
110	145	4	—	7	1	117	141	83 200	578 000	2 000
120	155	4	—	7	1	127	151	87 900	634 000	1 800
130	170	5	—	9	1	137	165	120 000	839 000	1 700

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

THRUST BEARINGS

Thrust Roller Bearings





AZ AZK WS GS

Shaft dia. 10 — 65mm

Shaft dia. mm	Identification number: Thrust roller bearing	Mass (Ref.) g	Thrust roller bearing	Mass (Ref.) g	Inner ring	Outer ring	Mass (Ref.) g
10	AZ 10249	24.6	AZK 10243.5	8.6	WS 1024	GS 1024	8
12	AZ 12269	26.5	AZK 12263.5	8.7	WS 1226	GS 1226	8.9
15	AZ 15289	28	AZK 15283.5	9.4	WS 1528	GS 1528	9.3
17	AZ 17309	30.5	AZK 17303.5	10.1	WS 1730	GS 1730	10.2
20	AZ 203510	45.5	AZK 20354.5	17.9	WS 2035	GS 2035	13.8
25	AZ 254211	70	AZK 25425	28	WS 2542	GS 2542	21
30	AZ 304711	79	AZK 30475	31	WS 3047	GS 3047	24
	AZ 305216	160	AZK 30527.5	70	WS 3052	GS 3052	45
35	AZ 355212	99	AZK 35525	36	WS 3552	GS 3552	31.5
	AZ 356218	260	AZK 35627.5	98	WS 3562	GS 3562	81
40	AZ 406013	139	AZK 40606	54	WS 4060	GS 4060	42.5
	AZ 406819	310	AZK 40689	132	WS 4068	GS 4068	89
45	AZ 456514	169	AZK 45656	62	WS 4565	GS 4565	53.5
	AZ 457320	360	AZK 45739	144	WS 4573	GS 4573	108
50	AZ 507014	185	AZK 50706	68	WS 5070	GS 5070	58.5
	AZ 507822	430	AZK 507811	194	WS 5078	GS 5078	118
55	AZ 557816	275	AZK 55786	89	WS 5578	GS 5578	93
	AZ 559025	725	AZK 559011	275	WS 5590	GS 5590	225
60	AZ 608517	345	AZK 60857.5	135	WS 6085	GS 6085	105
	AZ 609526	770	AZK 609511	290	WS 6095	GS 6095	240
	AZ 6013026	2 090	AZK 6013010	790	WS 60130	GS 60130	650
65	AZ 659018	380	AZK 65907.5	132	WS 6590	GS 6590	124
	AZ 6510027	860	AZK 6510011	310	WS 65100	GS 65100	275

Notes(1) Minimum allowable value of chamfer dimension *r*
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 25% of this value is allowable.

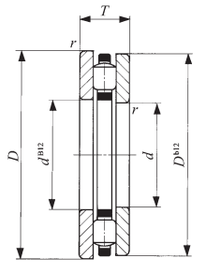
Boundary dimensions mm										Standard mounting dimensions mm		Basic dynamic load rating	Basic static load rating	Allowable rotational speed(2)
d	D	T	d_c	D_c	D_w	B	$r_{s\,min}$ (1)	C_i	C_e	d_a Min.	D_a Max.	C N	C_0 N	rpm
10	24	9	10.04	23.6	3.5	2.75	0.3	13	21	21	13	8 990	19 100	18 000
12	26	9	12.04	25.6	3.5	2.75	0.3	15	23	23	16	10 400	23 900	16 000
15	28	9	15.04	27.6	3.5	2.75	0.3	17	25	25	18	10 200	23 900	14 000
17	30	9	17.04	29.6	3.5	2.75	0.3	19	27	27	20	11 400	28 600	13 000
20	35	10	20.04	34.6	4.5	2.75	0.3	22	33	33	23	19 000	48 700	11 000
25	42	11	25.05	41.6	5	3	0.6	28	39	39	28	22 700	60 700	9 000
30	47	11	30.05	46.5	5	3	0.6	33	44	44	33	27 400	81 000	8 000
30	52	16	30.05	51.5	7.5	4.25	0.6	35	49	48	36	38 400	95 700	7 500
35	52	12	35.05	51.5	5	3.5	0.6	38	49	49	39	29 100	91 100	7 000
35	62	18	35.05	61.5	7.5	5.25	1	42	58	57	43	47 900	135 000	6 500
40	60	13	40.05	59.5	6	3.5	0.6	44	57	57	44	41 700	133 000	6 000
40	68	19	40.05	67.5	9	5	1	45	64	64	46	68 700	195 000	5 500
45	65	14	45.05	64.5	6	4	0.6	49	62	62	49	40 800	133 000	5 500
45	73	20	45.05	72.5	9	5.5	1	50	69	69	51	75 700	227 000	5 000
50	70	14	50.05	69.5	6	4	0.6	54	67	67	54	43 300	148 000	5 000
50	78	22	50.05	77.5	11	5.5	1	55	74	73	56	84 300	232 000	4 500
55	78	16	55.05	77.5	6	5	0.6	59	75	75	60	51 700	192 000	4 500
55	90	25	55.05	89.5	11	7	1	63	85	84	63	108 000	332 000	4 000
60	85	17	60.05	84.5	7.5	4.75	1	65	81	81	66	64 600	224 000	4 000
60	95	26	60.05	94.5	11	7.5	1	68	90	89	68	106 000	332 000	4 000
60	130	26	60.05	129.5	10	8	1.5	79	119	119	80	158 000	634 000	3 000
65	90	18	65.05	89.5	7.5	5.25	1	70	86	86	71	68 300	247 000	4 000
65	100	27	65.05	99.5	11	8	1	73	95	94	73	116 000	379 000	3 500

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

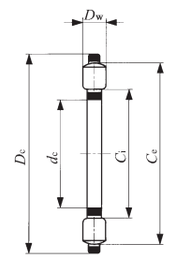
THRUST BEARINGS

Thrust Roller Bearings

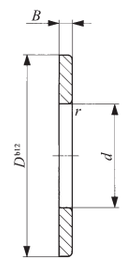




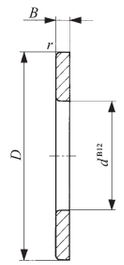
AZ



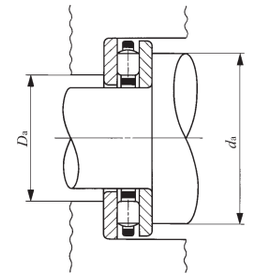
AZK



WS



GS



Shaft dia. 70 — 130mm

Shaft dia. mm	Identification number: Thrust roller bearing	Mass (Ref.) g	Thrust roller bearing	Mass (Ref.) g	Inner ring	Outer ring	Mass (Ref.) g
70	AZ 709518	420	AZK 70957.5	156	WS 7095	GS 7095	132
	AZ 7010527	905	AZK 7010511	325	WS 70105	GS 70105	290
	AZ 7014026	2 250	AZK 7014010	890	WS 70140	GS 70140	680
75	AZ 7510019	465	AZK 751007.5	159	WS 75100	GS 75100	153
	AZ 7511027	960	AZK 7511011	340	WS 75110	GS 75110	310
80	AZ 8010519	495	AZK 801057.5	171	WS 80105	GS 80105	162
	AZ 8011528	1 060	AZK 8011511	370	WS 80115	GS 80115	345
	AZ 8015026	2 500	AZK 8015010	920	WS 80150	GS 80150	790
85	AZ 8511019	530	AZK 851107.5	190	WS 85110	GS 85110	170
	AZ 8512531	1 460	AZK 8512512	510	WS 85125	GS 85125	475
90	AZ 9012022	790	AZK 901209	290	WS 90120	GS 90120	250
	AZ 9013535	2 040	AZK 9013514	750	WS 90135	GS 90135	645
	AZ 9016026	2 710	AZK 9016010	1 000	WS 90160	GS 90160	855
100	AZ 10013525	1 190	AZK 10013511	490	WS 100135	GS 100135	350
	AZ 10015038	2 720	AZK 10015015	980	WS 100150	GS 100150	870
	AZ 10019039	5 960	AZK 10019015	2 120	WS 100190	GS 100190	1 920
110	AZ 11014525	1 350	AZK 11014511	590	WS 110145	GS 110145	380
	AZ 11016040	3 220	AZK 11016017	1 320	WS 110160	GS 110160	950
	AZ 11020039	6 400	AZK 11020015	2 280	WS 110200	GS 110200	2 060
120	AZ 12015525	1 450	AZK 12015511	630	WS 120155	GS 120155	410
	AZ 12017542	4 020	AZK 12017518	1 640	WS 120175	GS 120175	1 190
	AZ 12022039	7 730	AZK 12022015	2 730	WS 120220	GS 120220	2 500
130	AZ 13017030	2 180	AZK 13017012	860	WS 130170	GS 130170	660
	AZ 13018542	4 300	AZK 13018518	1 760	WS 130185	GS 130185	1 270
	AZ 13023039	8 240	AZK 13023015	2 940	WS 130230	GS 130230	2 650

Shaft dia. mm	Boundary dimensions mm: d	D	T	d_c	D_c	D_w	B	$r_{s\,min}$ (1)	C_i	C_e	Standard mounting dimensions mm: d_a Min.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
70	70	95	18	70.05	94.5	7.5	5.25	1	75	91	91	76	72 000	269 000	3 500
	70	105	27	70.05	104.5	11	8	1	78	100	99	78	114 000	379 000	3 500
	70	140	26	70.05	139.5	10	8	1.1	89	129	129	90	169 000	713 000	3 000
75	75	100	19	75.05	99.5	7.5	5.75	1	80	96	96	81	71 100	269 000	3 500
	75	110	27	75.05	109.5	11	8	1	83	105	104	83	123 000	427 000	3 000
80	80	105	19	80.05	104.5	7.5	5.75	1	85	101	101	86	74 500	292 000	3 000
	80	115	28	80.05	114.5	11	8.5	1	88	110	109	88	122 000	427 000	3 000
	80	150	26	80.05	149.5	10	8	1.5	99	139	139	100	180 000	792 000	2 500
85	85	110	19	85.05	109.5	7.5	5.75	1	90	106	106	91	77 800	314 000	3 000
	85	125	31	85.05	124.5	12	9.5	1	95	119	118	95	145 000	513 000	3 000
90	90	120	22	90.05	119.5	9	6.5	1	97	116	115	97	99 700	390 000	3 000
	90	135	35	90.05	134.5	14	10.5	1.1	100	129	128	101	181 000	626 000	2 500
	90	160	26	90.05	159.5	10	8	1.5	109	149	149	110	189 000	871 000	2 500
100	100	135	25	100.05	134.5	11	7	1	108	130	129	108	136 000	522 000	2 500
	100	150	38	100.05	149.5	15	11.5	1.1	112	143	142	113	219 000	796 000	2 500
	100	190	39	100.1	189.3	15	12	1.5	119	179	177	120	333 000	1 420 000	2 000
110	110	145	25	110.1	144.5	11	7	1	118	140	139	118	142 000	569 000	2 500
	110	160	40	110.1	159.5	17	11.5	1.1	120	154	153	121	282 000	1 030 000	2 000
	110	200	39	110.1	199.3	15	12	2	129	188	187	130	388 000	1 770 000	2 000
120	120	155	25	120.1	154.5	11	7	1	128	150	149	128	149 000	617 000	2 000
	120	175	42	120.1	174.5	18	12	1.1	132	168	167	133	313 000	1 160 000	2 000
	120	220	39	120.1	219	15	12	2.1	141	207	206	142	415 000	1 980 000	1 800
130	130	170	30	130.1	169.5	12	9	1	140	164	163	140	176 000	741 000	2 000
	130	185	42	130.1	184.5	18	12	1.5	142	178	177	143	333 000	1 290 000	1 900
	130	230	39	130.1	229	15	12	2.1	151	217	216	152	440 000	2 180 000	1 700

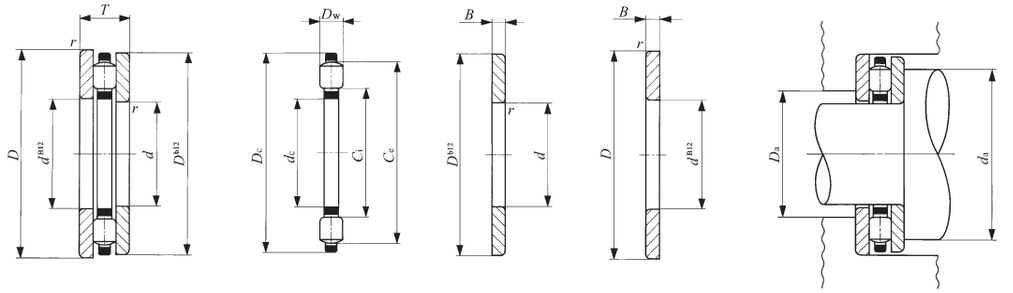
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 25% of this value is allowable.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

THRUST BEARINGS

Thrust Roller Bearings





AZ AZK WS GS

Shaft dia. 140 — 280mm

Shaft dia. mm	Identification number: Thrust roller bearing	Mass (Ref.) g	Thrust roller bearing	Mass (Ref.) g	Inner ring	Outer ring	Mass (Ref.) g
140	**AZ 14018031**	2 410	**AZK 14018012**	920	**WS 140180**	**GS 140180**	745
	AZ 14019542	4 560	**AZK 14019518**	1 860	**WS 140195**	**GS 140195**	1 350
	AZ 14024039	8 680	**AZK 14024015**	3 100	**WS 140240**	**GS 140240**	2 790
150	**AZ 15019031**	2 560	**AZK 15019012**	980	**WS 150190**	**GS 150190**	790
	AZ 15020542	4 840	**AZK 15020518**	1 980	**WS 150205**	**GS 150205**	1 430
	AZ 15025039	9 140	**AZK 15025015**	3 260	**WS 150250**	**GS 150250**	2 940
160	**AZ 16020031**	2 710	**AZK 16020012**	1 030	**WS 160200**	**GS 160200**	840
	AZ 16027039	10 800	**AZK 16027015**	3 840	**WS 160270**	**GS 160270**	3 480
170	**AZ 17023045**	6 220	**AZK 17023019**	2 420	**WS 170230**	**GS 170230**	1 900
	AZ 17028039	11 300	**AZK 17028015**	4 020	**WS 170280**	**GS 170280**	3 640
180	**AZ 18024045**	6 540	**AZK 18024019**	2 540	**WS 180240**	**GS 180240**	2 000
	AZ 18031039	14 600	**AZK 18031015**	5 200	**WS 180310**	**GS 180310**	4 700
190	**AZ 19025548**	8 060	**AZK 19025520**	3 100	**WS 190255**	**GS 190255**	2 480
	AZ 19032039	15 000	**AZK 19032015**	5 280	**WS 190320**	**GS 190320**	4 860
200	**AZ 20026548**	8 430	**AZK 20026520**	3 250	**WS 200265**	**GS 200265**	2 590
	AZ 20034039	17 200	**AZK 20034015**	6 120	**WS 200340**	**GS 200340**	5 540
220	**AZ 22029050**	10 400	**AZK 22029022**	4 280	**WS 220290**	**GS 220290**	3 060
	AZ 22036052	24 000	**AZK 22036020**	8 000	**WS 220360**	**GS 220360**	8 000
240	**AZ 24031554**	13 200	**AZK 24031524**	5 520	**WS 240315**	**GS 240315**	3 840
	AZ 24038052	26 500	**AZK 24038020**	9 440	**WS 240380**	**GS 240380**	8 530
260	**AZ 26034055**	15 400	**AZK 26034025**	6 600	**WS 260340**	**GS 260340**	4 400
	AZ 26042080	51 600	**AZK 26042030**	18 200	**WS 260420**	**GS 260420**	16 700
280	**AZ 28044080**	54 600	**AZK 28044030**	19 200	**WS 280440**	**GS 280440**	17 700

Boundary dimensions mm d	D	T	d_c	D_c	D_w	B	$r_{s\,min}$ (1)	C_i	C_e	Standard mounting dimensions mm d_a Min.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (2) rpm
140	180	31	140.1	179.5	12	9.5	1	150	174	173	150	184 000	798 000	1 900
140	195	42	140.1	194.5	18	12	1.5	152	188	187	153	353 000	1 420 000	1 800
140	240	39	140.1	239	15	12	2.1	161	227	226	162	435 000	2 180 000	1 600
150	190	31	150.1	189.5	12	9.5	1	160	184	183	160	181 000	798 000	1 800
150	205	42	150.1	204.5	18	12	1.5	162	198	197	163	349 000	1 420 000	1 700
150	250	39	150.1	249	15	12	2.1	171	237	236	172	459 000	2 380 000	1 500
160	200	31	160.1	199.5	12	9.5	1	170	194	193	170	189 000	855 000	1 700
160	270	39	160.1	269	15	12	3	183	256	255	184	519 000	2 850 000	1 400
170	230	45	170.1	229	19	13	1.5	183	221	220	184	406 000	1 730 000	1 500
170	280	39	170.1	279	15	12	3	193	266	265	194	543 000	3 070 000	1 300
180	240	45	180.1	239	19	13	1.5	193	231	230	194	426 000	1 870 000	1 400
180	310	39	180.1	308	15	12	3	204	294	293	205	619 000	3 710 000	1 200
190	255	48	190.1	254	20	14	2	205	245	244	206	470 000	2 080 000	1 300
190	320	39	190.1	318	15	12	4	214	304	303	215	647 000	3 980 000	1 200
200	265	48	200.15	264	20	14	2	215	255	254	216	465 000	2 080 000	1 300
200	340	39	200.15	338	15	12	4	227	323	322	228	710 000	4 580 000	1 100
220	290	50	220.15	289	22	14	2	236	280	278	237	557 000	2 530 000	1 300
220	360	52	220.15	358	20	16	4	246	343	342	247	943 000	5 520 000	1 000
240	315	54	240.15	314	24	15	2	256	304	302	257	695 000	3 250 000	1 100
240	380	52	240.15	378	20	16	4	266	363	362	267	977 000	5 910 000	1 000
260	340	55	260.15	339	25	15	2.1	278	328	326	279	739 000	3 510 000	1 000
260	420	80	260.15	418	30	25	5	289	402	400	291	1 430 000	7 490 000	900
280	440	80	280.15	438	30	25	5	309	422	420	311	1 420 000	7 490 000	800

Notes(1) Minimum allowable value of chamfer dimension *r*
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 25% of this value is allowable.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

COMBINED TYPE NEEDLE ROLLER BEARINGS

- Needle Roller Bearings with Thrust Ball Bearing
- Needle Roller Bearings with Thrust Roller Bearing
- Needle Roller Bearings with Angular Contact Ball Bearing
- Needle Roller Bearings with Three-point Contact Ball Bearing



Structure and Features

IKO Combined Type Needle Roller Bearings are combinations of a radial bearing and a thrust bearing. Caged needle roller bearings are used as radial bearings and thrust ball bearings or thrust roller bearings are used as thrust bearings. They are compact and very economical, and can be subjected to radial loads and axial loads simultaneously.
They are widely used for machine tools, textile machinery, and industrial machinery.

Types

In IKO Combined Type Needle Roller Bearings, the types shown in Table 1 are available.

Table 1.1 Type of bearing

Type	Combined with thrust ball bearing		Combined with thrust roller bearing	
	Without inner ring	With inner ring	Without inner ring	With inner ring
——	NAX	NAXI	NBX	NBXI
With dust cover	NAX···Z	NAXI···Z	NBX···Z	NBXI···Z

Table 1.2 Type of bearing

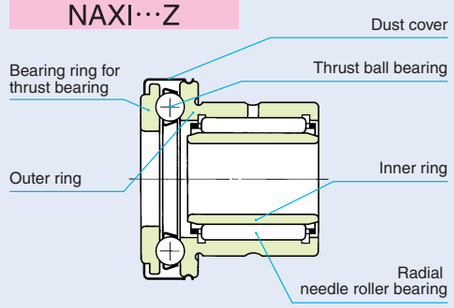
Type	Combined with angular contact ball bearing	Combined with three-point contact ball bearing
Model code	NATA	NATB

Needle Roller Bearings with Thrust Ball Bearing

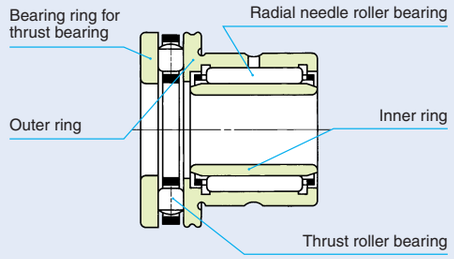
In this series, needle roller bearings are combined with thrust ball bearings to receive thrust loads.
In bearings with a dust cover, the dust cover is formed from a thin steel plate and fixed to a groove cut on the outer cylindrical surface of the outer ring collar. The cover forms a labyrinth with the thrust raceway ring, and is therefore effective in preventing leakage of grease and penetration of dust and dirt.
In the case of bearings without an inner ring, the tolerances of roller set bore diameter F_w are shown in Table 14 on page A33. Therefore, the required radial internal clearances can be selected by combining the bearings with shafts that have been heat-treated and finished by grinding as shown in Table 23 on page A42 and Table 26 on page A44.

Structures of Combined Type Needle Roller Bearings

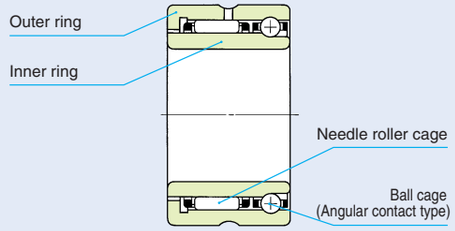
NAXI···Z



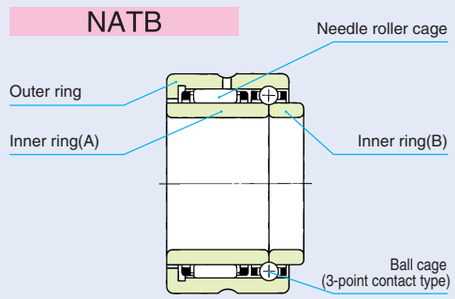
NBXI



NATA



NATB



Needle Roller Bearings with Thrust Roller Bearing

In this series, needle roller bearings are combined with thrust roller bearings to receive thrust loads.
Their axial load ratings are greater than those of bearings that are combined with thrust ball bearings. Also, elastic deformation of the rolling contact surfaces under load is minimal. Furthermore, the thrust bearing section is finished to high accuracy, and therefore high rotational accuracy is obtained in the case of both vertical and horizontal shafts.
Like the needle roller bearings with thrust ball bearing, this series also includes bearings with a dust cover and bearings with an inner ring.

Needle Roller Bearings with Angular Contact Ball Bearing

In this series, caged needle roller bearings are combined with angular contact ball bearings to receive thrust loads. These bearings conform to the international dimension series #59, which is based on the ISO Standard. They can withstand heavy radial loads and unidirectional axial loads simultaneously.
When the axial load exceeds 25% of the radial load, the radial load will be induced in the angular contact ball bearing, and bearing life will be affected. The relationship between the two loads must therefore be taken into careful consideration.

Needle Roller Bearings with Three-point Contact Ball Bearing

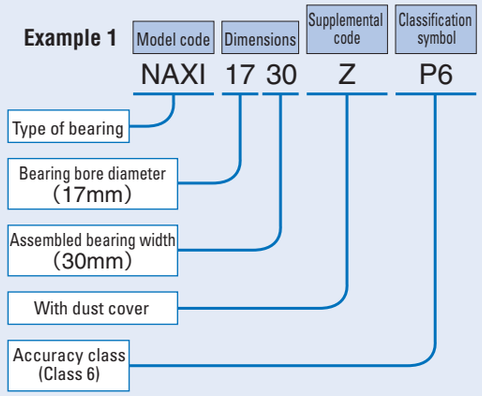
These bearings can withstand heavy radial loads and bi-directional axial loads at the same time during high-speed rotation.
Since the non-interchangeable inner rings are separated at the center of the ball raceway surface, they must be firmly tightened against the shaft in the axial direction. The axial clearance of this bearing is 0.1～0.3 mm, and like NATA59, the axial load should not exceed 25% of the radial load.

Identification Number

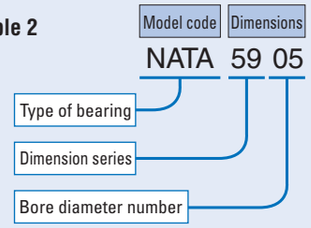
The identification number of Combined Type Needle Roller Bearings consists of a model code, dimensions, any supplemental codes and a classification symbol. Some examples are shown below.

Examples of identification number

Example 1



Example 2



Accuracy

Dimensional accuracy and rotational accuracy of Combined Type Needle Roller Bearings are based on Table 2 below and Tables 12 and 13 on page A31. Thickness variations of thrust rings of NAX(I) and NBX(I) are based on Table 2.4 on page F5.
Bore diameter of the small width inner ring of NATB59 is made for a transition fit with k5 tolerance shaft.

Table 2 Tolerances unit: mm

Item / Type of bearing	Dimension	Dimension symbol	Tolerance
NAX(I)(1) NBX(I)(1)	Bore dia. of bearing ring for thrust bearing	d_i	E7
	Assembled bearing width	L	0 −0.25
	Bearing height of thrust bearing	H	0 −0.20
NATB59	Width of inner ring	B	0 −0.3

Note(1) Also applicable to bearings with dust cover

Clearance

Combined Type Needle Roller Bearings are manufactured to have the radial internal clearance CN shown in Table 18 on page A37.

Fit

The recommended fits for Combined Type Needle Roller Bearings are shown in Table 3.

Table 3 Recommended fits

Item / Type of bearing	Tolerance class		
	Shaft		Housing bore
	Without inner ring	With inner ring	
NAX(I)(1) NBX(I)(1)	h5，k5	k5	K6，M6
NATA59 NATB59	——	k5(2)	M6(2)

Notes(1) The housing bore for the thrust bearing must be machined to be more than 0.5 mm larger than the outside diameters D_1 and D_2 to ensure that it does not incur radial loads.
(2) If the fit is made tighter than specified in this table, radial loads will act upon the thrust bearing, limiting its function.

Lubrication

Grease is not prepacked in Combined Type Needle Roller Bearings, so perform proper lubrication for use. Operating without lubrication will increase the wear of the rolling contact surfaces and shorten the bearing life.

Oil Hole

The outer ring of Combined Type Needle Roller Bearings has an oil groove and an oil hole. When outer rings with multiple oil holes or inner rings with oil hole(s) are required, please contact IKO.

Rating Life

In Combined Type Needle Roller Bearings, caged needle roller bearings are subjected to radial loads while thrust bearings receive axial loads. Therefore, it is necessary to calculate their lives respectively (page A17).

Mounting

Fig.1 shows mounting examples of Combined Type Needle Roller Bearings. When applying preload to the NAX and NBX models, it is recommended that thrust raceway rings are not tightened directly with nuts, but are tightened using springs as shown in Fig. 2.
Mounting two NATA models symmetrically allows them to be subjected to two-way axial loads. When mounting these models, an axial clearance of 0.2～0.3 mm should be provided in the angular contact ball bearings so that radial loads are not applied to the angular contact ball bearings.
Dimensions related to mounting should be based on the table of dimensions.

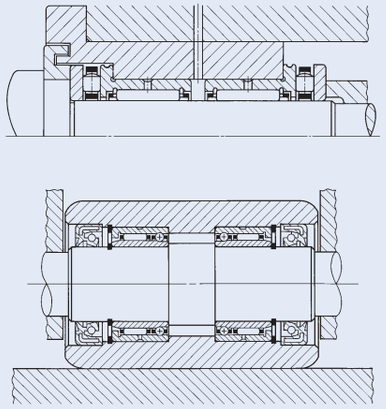


Fig.1 Mounting examples



Fig.2 Mounting example when applying preload

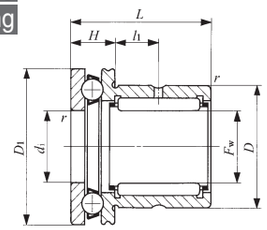
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

COMBINED TYPE NEEDLE ROLLER BEARINGS

Needle Roller Bearings with Thrust Ball Bearing — Without Inner Ring
Needle Roller Bearings with Thrust Roller Bearing — Without Inner Ring





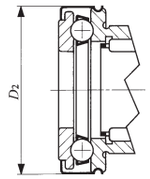


NAX

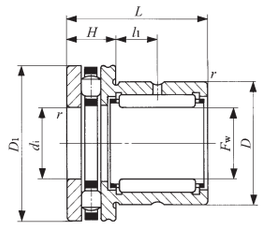
Shaft dia. 10 — 70mm

Shaft dia. mm	Identification number	Mass (Ref.) g	With dust cover	Mass (Ref.) g		Mass (Ref.) g	With dust cover	Mass (Ref.) g
10	**NAX 1023**	38.5	**NAX 1023Z**	40	—	—	—	—
12	**NAX 1223**	43.5	**NAX 1223Z**	45.5	—	—	—	—
15	**NAX 1523**	47.5	**NAX 1523Z**	48.5	—	—	—	—
	—	—	—	—	**NBX 1523**	54	**NBX 1523Z**	55
17	**NAX 1725**	54	**NAX 1725Z**	56	—	—	—	—
	—	—	—	—	**NBX 1725**	61	**NBX 1725Z**	63
20	**NAX 2030**	85.5	**NAX 2030Z**	89	—	—	—	—
	—	—	—	—	**NBX 2030**	94	**NBX 2030Z**	97.5
25	**NAX 2530**	131	**NAX 2530Z**	135	—	—	—	—
	—	—	—	—	**NBX 2530**	143	**NBX 2530Z**	147
30	**NAX 3030**	145	**NAX 3030Z**	151	—	—	—	—
	—	—	—	—	**NBX 3030**	160	**NBX 3030Z**	166
35	**NAX 3530**	169	**NAX 3530Z**	176	—	—	—	—
	—	—	—	—	**NBX 3530**	186	**NBX 3530Z**	193
40	**NAX 4032**	219	**NAX 4032Z**	227	—	—	—	—
	—	—	—	—	**NBX 4032**	240	**NBX 4032Z**	248
45	**NAX 4532**	264	**NAX 4532Z**	273	—	—	—	—
	—	—	—	—	**NBX 4532**	293	**NBX 4532Z**	302
50	**NAX 5035**	287	**NAX 5035Z**	297	—	—	—	—
	—	—	—	—	**NBX 5035**	315	**NBX 5035Z**	325
60	**NAX 6040**	417	**NAX 6040Z**	454	—	—	—	—
	—	—	—	—	**NBX 6040**	501	**NBX 6040Z**	538
70	**NAX 7040**	555	**NAX 7040Z**	606	—	—		—

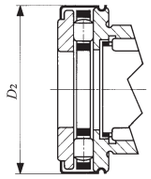
Notes(1) Minimum allowable value of chamfer dimension *r*
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 70% of this value is allowable in the NAX series, and a maximum of 25% of this value is allowable in the NBX series.
Remarks1. The outer ring has an oil groove and an oil hole.
2. Grease is not prepacked. Perform proper lubrication.



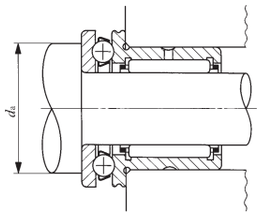
NAX···Z



NBX



NBX···Z



Boundary dimensions mm									Standard mounting dimension d_a	Basic dynamic load rating C		Basic static load rating C_0		Allowable rotational speed(2)
F_w	D	D_1	D_2	L	H	l_1	$r_{s\,min}$ (1)	d_i	Min. mm	Radial N	Axial N	Radial N	Axial N	rpm
10	19	24	25	23	9	6.5	0.3	10	18	8 230	10 000	9 190	11 100	9 500
12	21	26	27	23	9	6.5	0.3	12	20	9 250	9 670	11 200	11 100	9 000
15	24	28	29	23	9	6.5	0.3	15	23	12 300	9 930	14 900	12 200	8 500
15	24	28	29	23	9	6.5	0.3	15	26	12 300	10 200	14 900	23 900	14 000
17	26	30	31	25	9	8	0.3	17	25	12 900	10 800	16 300	14 500	8 500
17	26	30	31	25	9	8	0.3	17	28	12 900	11 400	16 300	28 600	13 000
20	30	35	36	30	10	10.5	0.3	20	29	17 600	14 200	25 400	19 700	7 500
20	30	35	36	30	10	10.5	0.3	20	33	17 600	19 000	25 400	48 700	11 000
25	37	42	43	30	11	9.5	0.6	25	35	20 000	19 600	32 100	29 700	7 000
25	37	42	43	30	11	9.5	0.6	25	40	20 000	22 700	32 100	60 700	9 000
30	42	47	48	30	11	9.5	0.6	30	40	25 100	20 400	40 100	33 600	6 500
30	42	47	48	30	11	9.5	0.6	30	45	25 100	27 400	40 100	81 000	8 000
35	47	52	53	30	12	9	0.6	35	45	26 900	21 200	46 200	37 600	6 000
35	47	52	53	30	12	9	0.6	35	50	26 900	29 100	46 200	91 100	7 000
40	52	60	61	32	13	10	0.6	40	52	29 400	26 900	54 100	50 000	5 500
40	52	60	61	32	13	10	0.6	40	57	29 400	41 700	54 100	133 000	6 000
45	58	65	66.5	32	14	9	0.6	45	57	31 000	27 900	60 200	55 100	5 000
45	58	65	66.5	32	14	9	0.6	45	62	31 000	40 800	60 200	133 000	5 500
50	62	70	71.5	35	14	10	0.6	50	62	42 200	28 800	83 400	60 100	4 500
50	62	70	71.5	35	14	10	0.6	50	67	42 200	43 300	83 400	148 000	5 000
60	72	85	86.5	40	17	12	1	60	75	47 500	41 400	103 000	89 700	4 000
60	72	85	86.5	40	17	12	1	60	82	47 500	64 600	103 000	224 000	4 000
70	85	95	96.5	40	18	11	1	70	85	55 500	43 100	120 000	101 000	3 500

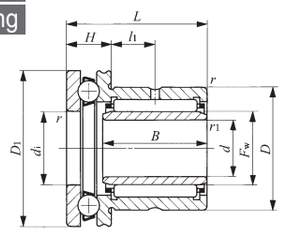
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

COMBINED TYPE NEEDLE ROLLER BEARINGS

Needle Roller Bearings with Thrust Ball Bearing — With Inner Ring
Needle Roller Bearings with Thrust Roller Bearing — With Inner Ring





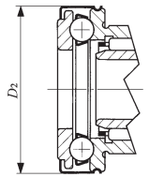


NAXI

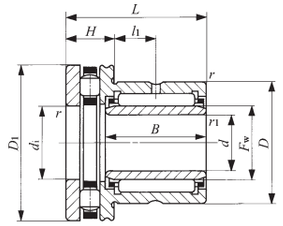
Shaft dia. 7 — 60mm

Shaft dia. mm	Identification number	Mass (Ref.) g	With dust cover	Mass (Ref.) g		Mass (Ref.) g	With dust cover	Mass (Ref.) g	d	D	D_1
7	**NAXI 723**	43.5	**NAXI 723Z**	45	—	—	—	—	7	19	24
9	**NAXI 923**	49.5	**NAXI 923Z**	51.5	—	—	—	—	9	21	26
12	**NAXI 1223**	55.5	**NAXI 1223Z**	56.5	—	—	—	—	12	24	28
	—	—	—	—	**NBXI 1223**	62	**NBXI 1223Z**	63	12	24	28
14	**NAXI 1425**	63.5	**NAXI 1425Z**	65.5	—	—	—	—	14	26	30
	—	—	—	—	**NBXI 1425**	70.5	**NBXI 1425Z**	72.5	14	26	30
17	**NAXI 1730**	99	**NAXI 1730Z**	103	—	—	—	—	17	30	35
	—	—	—	—	**NBXI 1730**	108	**NBXI 1730Z**	111	17	30	35
20	**NAXI 2030**	159	**NAXI 2030Z**	163	—	—	—	—	20	37	42
	—	—	—	—	**NBXI 2030**	171	**NBXI 2030Z**	175	20	37	42
25	**NAXI 2530**	179	**NAXI 2530Z**	185	—	—	—	—	25	42	47
	—	—	—	—	**NBXI 2530**	194	**NBXI 2530Z**	200	25	42	47
30	**NAXI 3030**	208	**NAXI 3030Z**	215	—	—	—	—	30	47	52
	—	—	—	—	**NBXI 3030**	225	**NBXI 3030Z**	232	30	47	52
35	**NAXI 3532**	265	**NAXI 3532Z**	273	—	—	—	—	35	52	60
	—	—	—	—	**NBXI 3532**	286	**NBXI 3532Z**	294	35	52	60
40	**NAXI 4032**	315	**NAXI 4032Z**	324	—	—	—	—	40	58	65
	—	—	—	—	**NBXI 4032**	344	**NBXI 4032Z**	353	40	58	65
45	**NAXI 4535**	358	**NAXI 4535Z**	368	—	—	—	—	45	62	70
	—	—	—	—	**NBXI 4535**	386	**NBXI 4535Z**	396	45	62	70
50	**NAXI 5040**	582	**NAXI 5040Z**	619	—	—	—	—	50	72	85
	—	—	—	—	**NBXI 5040**	666	**NBXI 5040Z**	703	50	72	85
60	**NAXI 6040**	750	**NAXI 6040Z**	801	—	—	—	—	60	85	95

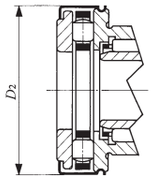
Notes(1) Minimum allowable value of chamfer dimension r or r_1
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 70% of this value is allowable in the NAXI series, and a maximum of 25% of this value is allowable in the NBXI series.
Remarks1. The outer ring has an oil groove and an oil hole.
2. Grease is not prepacked. Perform proper lubrication.



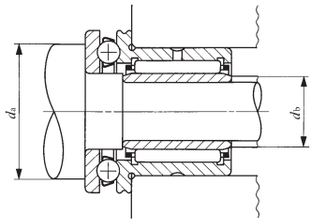
NAXI···Z



NBXI



NBXI···Z



Boundary dimensions mm D_2	L	B	H	l_1	$r_{s\,min}$ (1)	$r_{1s\,min}$ (1)	F_w	d_i	Standard mounting dimensions mm d_a Min.	d_b	Basic dynamic load rating C Radial N	Axial N	Basic static load rating C_0 Radial N	Axial N	Allowable rotational speed(2) rpm	Assembled inner ring
25	23	16	9	6.5	0.3	0.2	10	10	18	9	8 230	10 000	9 190	11 100	9 500	**LRT 71016**
27	23	16	9	6.5	0.3	0.3	12	12	20	11	9 250	9 670	11 200	11 100	9 000	**LRT 91216**
29	23	16.5	9	6.5	0.3	0.3	15	15	23	14	12 300	9 930	14 900	12 200	8 500	**LRT 121516**
29	23	16.5	9	6.5	0.3	0.3	15	15	26	14	12 300	10 200	14 900	23 900	14 000	**LRT 121516**
31	25	17	9	8	0.3	0.3	17	17	25	16	12 900	10 800	16 300	14 500	8 500	**LRT 141717**
31	25	17	9	8	0.3	0.3	17	17	28	16	12 900	11 400	16 300	28 600	13 000	**LRT 141717**
36	30	20.5	10	10.5	0.3	0.3	20	20	29	19	17 600	14 200	25 400	19 700	7 500	**LRT 172020**
36	30	20.5	10	10.5	0.3	0.3	20	20	33	19	17 600	19 000	25 400	48 700	11 000	**LRT 172020**
43	30	20.5	11	9.5	0.6	0.3	25	25	35	24	20 000	19 600	32 100	29 700	7 000	**LRT 202520**
43	30	20.5	11	9.5	0.6	0.3	25	25	40	24	20 000	22 700	32 100	60 700	9 000	**LRT 202520**
48	30	20.5	11	9.5	0.6	0.3	30	30	40	29	25 100	20 400	40 100	33 600	6 500	**LRT 253020**
48	30	20.5	11	9.5	0.6	0.3	30	30	45	29	25 100	27 400	40 100	81 000	8 000	**LRT 253020**
53	30	20	12	9	0.6	0.3	35	35	45	34	26 900	21 200	46 200	37 600	6 000	**LRT 303520**
53	30	20	12	9	0.6	0.3	35	35	50	34	26 900	29 100	46 200	91 100	7 000	**LRT 303520**
61	32	20	13	10	0.6	0.3	40	40	52	39	29 400	26 900	54 100	50 000	5 500	**LRT 354020**
61	32	20	13	10	0.6	0.3	40	40	57	39	29 400	41 700	54 100	133 000	6 000	**LRT 354020**
66.5	32	20	14	9	0.6	0.3	45	45	57	44	31 000	27 900	60 200	55 100	5 000	**LRT 404520**
66.5	32	20	14	9	0.6	0.3	45	45	62	44	31 000	40 800	60 200	133 000	5 500	**LRT 404520**
71.5	35	25	14	10	0.6	0.3	50	50	62	49	42 200	28 800	83 400	60 100	4 500	**LRT 455025**
71.5	35	25	14	10	0.6	0.3	50	50	67	49	42 200	43 300	83 400	148 000	5 000	**LRT 455025**
86.5	40	25.5	17	12	1	1	60	60	75	59	47 500	41 400	103 000	89 700	4 000	**LRT 506025**
86.5	40	25.5	17	12	1	1	60	60	82	59	47 500	64 600	103 000	224 000	4 000	**LRT 506025**
96.5	40	25.5	18	11	1	1	70	70	85	68	55 500	43 100	120 000	101 000	3 500	**LRT 607025**

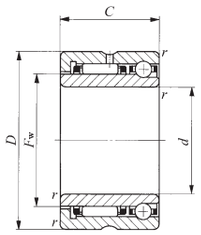
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

COMBINED TYPE NEEDLE ROLLER BEARINGS

Needle Roller Bearings with Angular Contact Ball Bearing
Needle Roller Bearings with Three-point Contact Ball Bearing





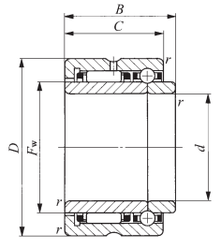


NATA59

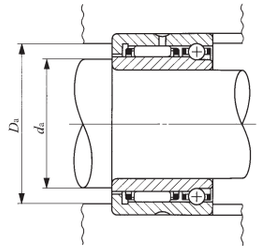
Shaft dia. 15 — 70mm

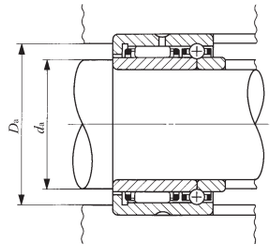
Shaft dia. mm	Identification number: Angular contact type	Mass (Ref.) g	Three-point contact type	Mass (Ref.) g	Boundary dimensions mm: d	D	C	B	$r_{s\,min}$(1)	F_w
15	**NATA 5902**	50.5	**NATB 5902**	53	15	28	18	20	0.3	20
17	**NATA 5903**	55.5	**NATB 5903**	58.5	17	30	18	20	0.3	22
20	**NATA 5904**	111	**NATB 5904**	115	20	37	23	25	0.3	25
25	**NATA 5905**	131	**NATB 5905**	136	25	42	23	25	0.3	30
30	**NATA 5906**	151	**NATB 5906**	157	30	47	23	25	0.3	35
35	**NATA 5907**	250	**NATB 5907**	260	35	55	27	30	0.6	42
40	**NATA 5908**	355	**NATB 5908**	375	40	62	30	34	0.6	48
45	**NATA 5909**	410	**NATB 5909**	435	45	68	30	34	0.6	55
50	**NATA 5910**	420	**NATB 5910**	445	50	72	30	34	0.6	58
55	**NATA 5911**	585	**NATB 5911**	615	55	80	34	38	1	63
60	**NATA 5912**	625	**NATB 5912**	660	60	85	34	38	1	68
65	**NATA 5913**	665	**NATB 5913**	710	65	90	34	38	1	75
70	**NATA 5914**	1 070	**NATB 5914**	1 130	70	100	40	45	1	80

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. Grease is not prepacked. Perform proper lubrication.



NATB59





Standard mounting dimensions mm: d_a Min.	D_a Max.	Basic dynamic load rating C: Radial N	Axial N	Basic static load rating C_0: Radial N	Axial N	Allowable rotational speed(2) rpm
17	26	7 710	1 900	10 200	2 920	20 000
19	28	8 220	2 050	11 500	3 340	18 000
22	35	14 300	3 810	18 400	6 110	16 000
27	40	15 800	4 300	22 100	7 520	13 000
32	45	17 700	4 550	26 800	8 460	11 000
39	51	24 000	4 890	42 100	9 870	9 500
44	58	30 600	5 350	60 400	11 800	8 500
49	64	32 600	5 450	68 500	12 700	7 000
54	68	33 600	5 660	72 500	13 600	7 000
60	75	39 500	10 400	74 400	24 700	6 500
65	80	41 800	10 700	82 200	26 700	6 000
70	85	43 800	11 000	90 200	28 700	5 500
75	95	56 400	13 500	127 000	35 000	5 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

INNER RINGS

●Inner Rings for Shell Type Needle Roller Bearings
●Inner Rings for General Usage



Structure and Features

IKO Inner Rings are heat-treated and finished by grinding to a high degree of accuracy. In the case of needle roller bearings, normally, the shafts are heat-treated and finished by grinding, and used as the raceway surfaces. However, when it is impossible to make shaft surfaces according to the specified surface hardness or surface roughness, inner rings are used.
Inner rings include those for Shell Type Needle Roller Bearings and those for general use and are available in a variety of dimensions. When shafts move axially or seals are used adjacent to bearings, wide inner rings can be selected.
Inner rings can also be used economically as bushings without requiring any additional machining.

Types

For Inner Rings, the types shown in Table 1 are available.

Table 1.1 Inner Rings for Shell Type Needle Roller Bearings

Series		Model codes of assembled bearings
Metric series	IRT	TA···Z, TLA···Z TAM, TLAM, YT, YTL
Inch series	IRB	BA···Z, BHA···Z BAM, BHAM, YB, YBH

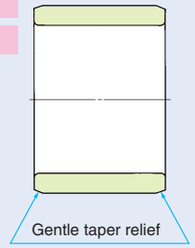
Remark For Inner Rings for Shell Type Needle Roller Bearings with Seal, please consult IKO.

Table 1.2 Inner Rings for General Usage

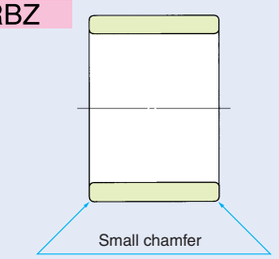
Series		Model codes of assembled bearings
Metric series	LRT	RNA 49, RNA 69 RNA 48, TAF, TR RNAF, NAX, NBX
	LRTZ	RNA 49···UU, RNA 69···UU GTR
Inch series	LRB	BR
	LRBZ···B	BR···UU
	LRBZ	GBR, GBR···UU

Shapes of Inner Rings

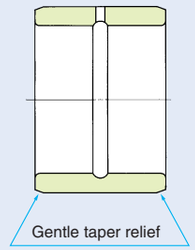
IRT
IRB
LRT



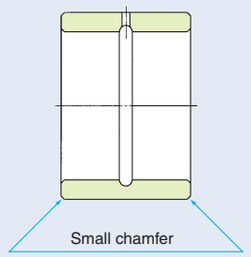
LRTZ
LRBZ



LRB



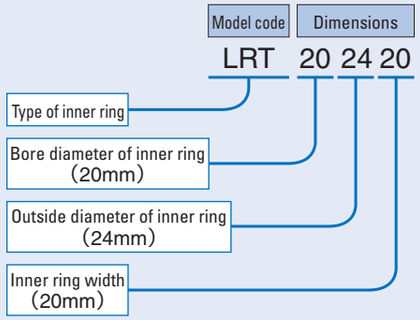
LRBZ···B



Identification number

The identification number of Inner Rings consists of a model code and dimensions. An example is shown below.

Example of identification number



Accuracy

Dimensional accuracy of Inner Rings is based on Table 2.

Inner Rings for Shell Type Needle Roller Bearings are manufactured so that exact radial internal clearances can be obtained when assembled with Shell Type Needle Roller Bearings. Inner Rings for General Usage produce CN clearance when used in the assembled bearings shown in Table 1.2. LRB and LRBZ···B models produce the radial internal clearances shown in Table 4 on page D5.

When clearances other than CN clearance or accuracy other than Class 0 are required, please consult IKO.

Table 2 Tolerances for inner ring

Model code	Tolerance
IRT LRT、LRTZ LRBZ	JIS Class 0 (See the table 12, page A31)
IRB	Based on Table 3
LRB LRBZ···B	Based on Table 4

Remark Tolerances of outside diameter of inner ring are based on Table 5.

Table 3 Tolerances of IRB

d Nominal inside diameter of inner ring mm		Δ_{dmp} Single plane mean bore diameter deviation		Δ_{Bs} Deviation of a single inner ring width		K_{ia} Radial runout of assembled bearing inner ring
Over	Incl.	High	Low	High	Low	Max.
2.5	10	0	−13	0	−250	10
10	18	0	−13	0	−250	10
18	30	0	−13	0	−250	13
30	50	0	−13	0	−250	15
50	80	0	−13	0	−250	20

Table 4 Tolerances of LRB,LRBZ···B

d Nominal inside diameter of inner ring mm		Δ_{dmp} Single plane mean bore diameter deviation		Δ_{Bs} Deviation of a single inner ring width		K_{ia} Radial runout of assembled bearing inner ring
Over	Incl.	High	Low	High	Low	Max.
—	19.050	0	−10	0	−130	10
19.050	30.162	0	−13	0	−130	13
30.162	50.800	0	−13	0	−130	15
50.800	82.550	0	−15	0	−130	20
82.550	120.650	0	−20	0	−130	25

Table 5 Tolerances of outside diameter of inner ring unit: μm

Model code	Tolerance
IRT	g5
IRB	0～−13
LRT，LRTZ，LRBZ	Based on Table 6
LRB，LRBZ···B	Based on Table 7

Table 7 Tolerances of outside diameters of LRB and LRBZ···B unit: μm

Nominal outside diameter of inner ring mm		Tolerance	
Over	Incl.	High	Low
—	18.034	−13	−23
18.034	25.908	−18	−30
25.908	30.226	−23	−36
30.226	35.052	−23	−38
35.052	50.038	−25	−41
50.038	80.010	−28	−46
80.010	100.076	−32	−56
100.076	102.108	−37	−66

Fit

The recommended fits between Inner Rings and shafts are shown in Table 22 on page A42.

Oil Hole

The number of oil holes is shown in Table 8.

When Inner Rings with an oil hole are especially required for a model without an oil hole, attach an "OH" to the end of the identification number when ordering.

Example: LRT 202420 OH

For Inner Rings with multiple oil holes, please consult IKO.

Table 8 Number of oil holes

Bearing type			Bore diameter of inner ring d mm	Number of oil holes
For Shell Type Needle Roller Bearings	Metric series	IRT		0
	Inch series	IRB		0
For General Usage	Metric series	LRT		0
		LRTZ		0
	Inch series	LRB	$d \leqq 76.200$	1
			$76.200 < d$	2
		LRBZ···B		1
		LRBZ		0

Remark Inner rings with an oil hole are provided with an oil groove.

H
IRT
IRB
LRT
LRB

Table 6 Tolerances of outside diameters for LRT, LRTZ and LRBZ (When the clearance is CN clearance) unit: μm

d Bore diameter of inner ring mm		F Outside diameter of inner ring mm																						d Bore diameter of inner ring mm	
		$3 < F \leqq 6$		$6 < F \leqq 10$		$10 < F \leqq 18$		$18 < F \leqq 30$		$30 < F \leqq 50$		$50 < F \leqq 80$		$80 < F \leqq 120$		$120 < F \leqq 180$		$180 < F \leqq 250$		$250 < F \leqq 315$		$315 < F \leqq 400$		$400 < F \leqq 500$			
Over	Incl.	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	Over	Incl.
—	24	−10	−27	−7	−23	−4	−18	0	−12																	—	24
24	30							0	−12	+5	−4															24	30
30	40									0	−9															30	40
40	50									−5	−19	0	−11													40	50
50	65											−10	−21													50	65
65	80											−10	−26	−4	−17											65	80
80	100													−14	−27											80	100
100	120													−14	−32	−7	−22									100	120
120	140															−17	−37									120	140
140	160															−27	−52									140	160
160	180																	−25	−46							160	180
180	200																	−40	−66							180	200
200	225																	−55	−86							200	225
225	250																			−54	−87					225	250
250	280																			−69	−107					250	280
280	315																					−68	−107			280	315
315	355																					−83	−127			315	355
355	400																					−128	−182	−122	−172	355	400
400	450																							−142	−202	400	450
450	500																							−152	−222	450	500